* NOTE:
1. VALUES MARKED * ARE ESTIMATES — VERIFY WITH DATASHEET / ON SITE.
2. DG SET / CHANGEOVER: NONE CONSIDERED. CONFIRM ON SITE; IF A DG EXISTS, ADD CHANGEOVER + INTERLOCK (SOLAR OFF ON DG).

CABLE DETAILS

CABLE SIZE	CURRENT RATING	CABLE CONNECTION	CABLE TYPE
1.1 KV, 1R X 3.5C X 95 SQ MM	216 A **	INVERTER TO ACDB	AL. XLPE INSULATED ARMOURED CABLE
1.1 KV, 1R X 3.5C X 95 SQ MM	216 A **	ACDB TO CLIENT LT PANEL	AL. XLPE INSULATED ARMOURED CABLE
1R X 1C X 4 SQ MM	55 A **	MODULE TO DCDB / INVERTER	XLPO UV Cu. CABLE FOR +VE & –VE
1R X 1C X 16 SQ MM	—	PV STRUCTURE TO EP-1	Cu EARTH CABLE
1R X 1C X 25 SQ MM	—	INVERTER / ACDB TO EP-2	Cu EARTH CABLE
1R X 1C X 50 SQ MM	—	LA TO EP-3	Cu EARTH CABLE

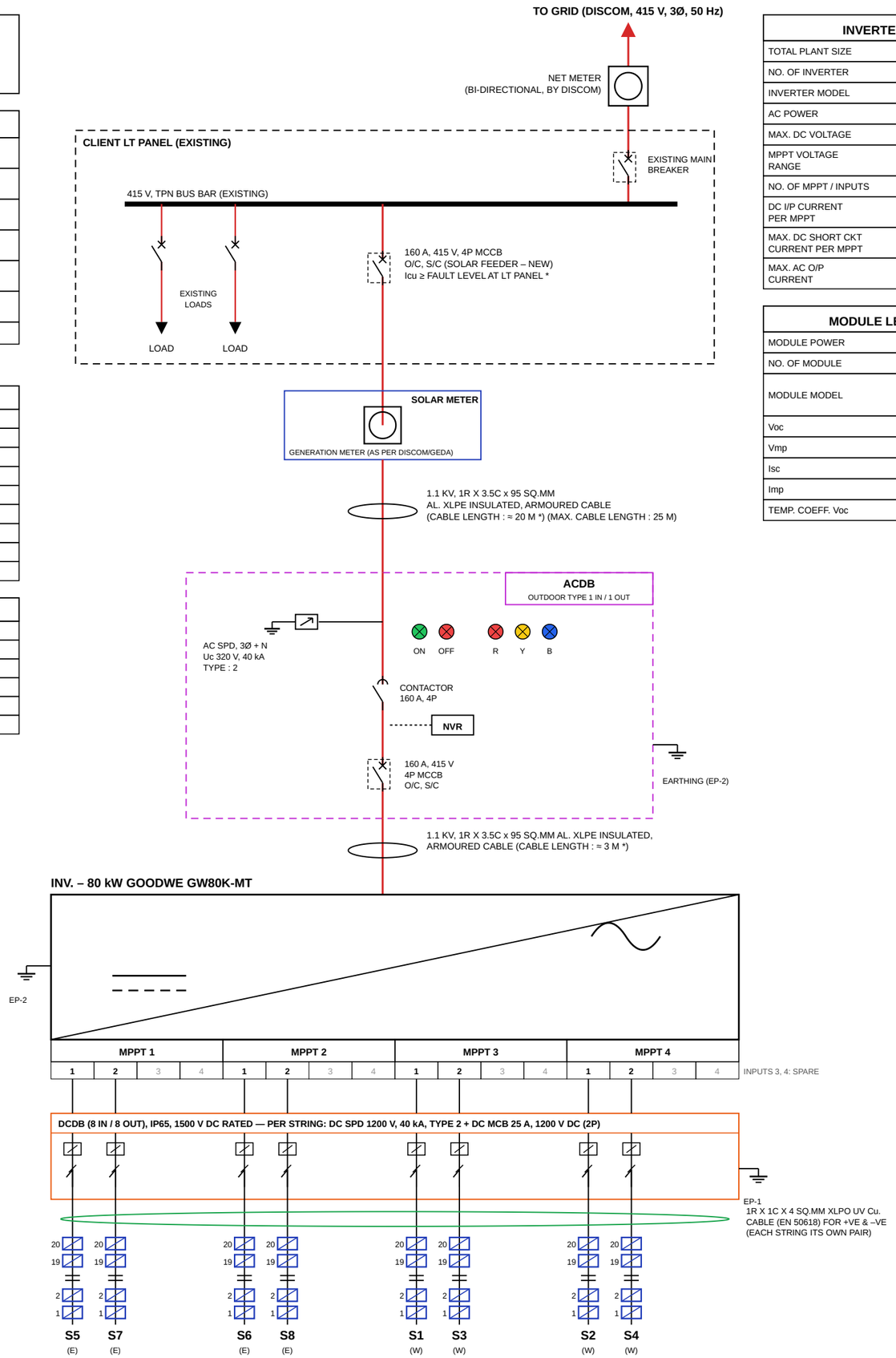
** CATALOGUE RATING IN AIR. DESIGN CURRENT: AC 1.25 x 127 A = 159 A; DC 1.25 x 15.96 A = 20 A PER STRING. CHECK WITH CABLE MAKER'S TABLE.

STRING / MPPT SCHEDULE

MPPT	INPUT	STRING	MODULES	SLOPE
1	1	S5	20	EAST
1	2	S7	20	EAST
2	1	S6	20	EAST
2	2	S8	20	EAST
3	1	S1	20	WEST
3	2	S3	20	WEST
4	1	S2	20	WEST
4	2	S4	20	WEST

DESIGN CHECK

Voc @ 5 °C (20 x 52.3 V)	1,046 V < 1,100 V ✓
Vmp @ 70 °C / STC / 5 °C	723 / 832 / 880 V (200–1000 V) ✓
CURRENT PER MPPT (Imp / Isc)	30.1 A ≤ 44 A / 31.9 A ≤ 55 A ✓
DC VOLTAGE DROP (4 SQ MM)	0.52 % – 1.03 % *
AC VOLTAGE DROP (95 SQ MM AL)	≈ 0.45 % *
STRING FUSE	NOT REQUIRED (≤ 2 STRINGS / MPPT)



INVERTER LEGEND

TOTAL PLANT SIZE	100.00 kWp
NO. OF INVERTER	1
INVERTER MODEL	GW80K-MT
AC POWER	80 kW
MAX. DC VOLTAGE	1100 V
MPPT VOLTAGE RANGE	200 – 1000 V
NO. OF MPPT / INPUTS	4 / 16
DC I/P CURRENT PER MPPT	44 A
MAX. DC SHORT CKT CURRENT PER MPPT	55 A
MAX. AC O/P CURRENT	127 A *

MODULE LEGEND (STC)

MODULE POWER	625 Wp
NO. OF MODULE	160
MODULE MODEL	ADANI AB-G12R-132-625 (N-TYPE TOPCon)
Voc	49.8 V *
Vmp	41.6 V *
Isc	15.96 A *
Imp	15.03 A *
TEMP. COEFF. Voc	–0.25 %/°C *

SYMBOLS NOMENCLATURE

	PV MODULE
	STRING INVERTER
	EARTHING
	SURGE PROTECTION DEVICE (SPD)
	MINIATURE CIRCUIT BREAKER (MCB)
	MOULDED CASE CIRCUIT BREAKER (MCCB)
	SOLAR ENERGY GENERATION METER
	NET METER (DISCOM)
	R, Y, B LIVE INDICATING LAMPS
	CIRCUIT BREAKER ON
	CIRCUIT BREAKER OFF
	CONTACTOR
NVR	NO VOLT RELAY
	CABLE

SYSTEM DESCRIPTION

SITE LOCATION	— (LAT, LONG TO BE ADDED)
DC CAPACITY	100.00 kWp
TOTAL PLANT CAPACITY (AC)	80 kW
DC / AC RATIO	1.25
MODULE CAPACITY	625 Wp (ADANI N-TOPCon)
MODULE DIMENSION (L X W X H)	2382 X 1134 X 30 mm *
TOTAL PV MODULES	160 Nos.
INVERTER CAPACITY	80 kW (GOODWE GW80K-MT) 1 No.
AZIMUTH / TILT	EAST-WEST (90° / 270°), ROOF SLOPE

INVERTER PROTECTION :- 80 kW (GOODWE GW80K-MT)
1. DC REVERSE POLARITY PROTECTION
2. AC SHORT CIRCUIT PROTECTION
3. INSULATION RESISTANCE DETECTION
4. RESIDUAL-CURRENT MONITORING UNIT (RCMU)
5. ANTI-ISLANDING PROTECTION
6. AC / DC SURGE PROTECTION TYPE II
7. DC SWITCH (INTEGRATED)
8. STRING CURRENT MONITORING
9. AC OVER / UNDER VOLTAGE & FREQUENCY PROTECTION
10. ANTI-PID / AFCI : OPTIONAL
11. SAFETY STANDARDS : AS PER GOODWE DATASHEET (IEC 62109-1/-2, IEC 61727, IEC 62116)

CUSTOMER SIGN AND STAMP:

REV.	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	APPROVED BY
R0	27-10-26	FIRST ISSUE			

CLIENT:	**JAIN GRANITE & MARBLE**
EPC CONTRACTOR:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara
DESIGN BY:	**NRG TECHNOLOGISTS PVT LTD** 989/6, GIDC Makarpura, Vadodara

DRAWING NO.: 01

DRAWING NAME: **SINGLE LINE DIAGRAM**

PROJECT TITLE: 100.00 kWp (DC)_ROOFTOP SOLAR PLANT

RELEASED FOR: ☐ PRELIMINARY ☐ TENDER ☐ REVISION ☐ FABRICATION ☐ CONSTRUCTION ■ APPROVAL

SCALE: **NTS**

DESIGN BY	DRAWN BY	CHECKED BY	APPROVED BY	PAPER SIZE	SHEET NO.
27-10-26	27-10-26	27-10-26	27-10-26	**A3**	**01 OF 01**